SIMONE ROMITI

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Computational physicist with a PhD and 4+ years of postdoctoral research experience in Monte Carlo simulations, Bayesian inference, and high-performance computing. Production-grade C++ and Python developer; main author and maintainer of open-source scientific libraries with CI/CD and versioned releases. Comfortable owning numerical methods end-to-end: derivation, implementation, statistical validation, and uncertainty quantification.

Resident in Switzerland with a B permit.

EDUCATION

PhD in Physics | title awarded in April 2022 — *Roma Tre University* | Nov. 2018 – Oct. 2021
M.S. in Physics | Grade: 110/110 *cum laude* | GPA: 3.98 / 4 — *Roma Tre University* | Oct. 2016 – Oct. 2018

B.S. in Physics | Grade: 110/110 *cum laude* | GPA: 3.85 / 4 — *Roma Tre University* | Oct. 2013 – Jul. 2016

WORK EXPERIENCE

University of Bern — Apr 2024 – Present
Postdoctoral Researcher | Software developer for scientific applications — Bern, Switzerland

University of Bonn — Nov 2021 – Mar 2024
Postdoctoral Researcher | Software developer for scientific applications — Bonn, Germany

COMPETENCES

- **Design and implementation of algorithms** for high-dimensional integrals with Markov Chain Monte Carlo, Langevin dynamics, Nested Sampling. Spectral methods for time series analysis.
- **Machine Learning applications to the solution of Partial Differential Equations** with Physics-Informed Neural Networks, convolutional networks, diffusion models, model validation.
- **Statistical analysis of terabyte-scale datasets** from data production to inference. Autocorrelation analysis, Akaike model averaging, and error budgeting.
- **Code optimization via parallel computing** using CPU and GPU offloading. Scaling-based optimizations for memory consumption and computational cost.
- **Scientific writing and reporting**: author of peer-reviewed papers in scientific journals, invited speaker at conferences and workshops, and organizer of scientific events.
- **Mentoring** and technical support for Master's and PhD students.

SOFTWARE TOOLS

Languages - Bash, C/C++, Python, R, SQL

Python libraries - PyTorch, NumPy, SciPy, Pandas, Matplotlib, Plotly, Streamlit, SymPy, Jupyter

HPC & Systems - CUDA, MPI, OpenMP, SLURM, EasyBuild

DevOps & Tools - Git, GitHub Actions, CI/CD, Docker, LaTeX, Quarto

Spoken Languages - Italian (native), English (proficient), German (basic)

ACHIEVEMENTS

- Developed novel algorithm using **Physics-Informed Neural Network** for solving the Schrödinger equation.
- **Principal Investigator** for $O(10^6)$ GPU-hour allocations on CSCS ALPS and GCS JUPITER 2026.
- Designed and operated **large-scale Monte Carlo simulations** on international HPC clusters, improving sampling efficiency.
- **Reduction of algorithmic complexity** for a class of multi-dimensional integrals from exponential to $O(N \log N)$ using parallelized Fast Fourier Transforms.
- Derived and implemented a **numerical method achieving machine-precision accuracy** for a class of constrained optimization problems.
- Developed a **Hybrid Monte Carlo + quantum-computing** method for estimating high-dimensional integrals.

SELECTED OPEN-SOURCE PROJECTS

- **lattice_data_tools** – Python toolkit for the production and statistical analysis of lattice simulation data. Main author and maintainer.
- **su2** – C++ templated library for Monte Carlo simulation of SU(N) theories with OpenMP parallelization. Main editor.

- **adiabatic_PINNs-suN** – Physics-Informed Neural Network solver for the Schrödinger equation in SU(N) lattice gauge theory; reference implementation for sole-author publication.
- **tmLQCD** – Large-scale C library for lattice simulations on GPU clusters (CUDA + MPI). Contributor.